

Instrument-Oriented Approach to Detecting and Representing Human Activity for Supporting Executive Functions and Learning*

ABSTRACT

The goal of this study is to develop a computer-interpretable model for activity detection and representation, based on existing informal models of how humans perform activity. Appropriate detection of purposeful human activity is an essential functionality of active assistive technology aiming at providing tailored support to individuals for improving activity performance and completion. The main contribution is the design of a model for detection and representation of human activities based on three categories of instruments, which is implemented as two generic and supplementary terminology models: an *event* ontology and a *core* ontology. The core ontology is extended for each new knowledge domain into a *domain* ontology. The model builds the base for personalization of services generated by the cooperative reasoning performed by a human collaborating with an intelligent and social software agent. Ongoing and future work includes user studies in the different application domains.

CCS CONCEPTS

•**Computer systems organization** → **Embedded systems**; *Redundancy*; Robotics; •**Networks** → Network reliability;

KEYWORDS

Ambient intelligence, Multi-agent system, Human-agent interaction, End-user development, Activity theory

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1 INTRODUCTION

An intelligent environment is typically dedicated to optimize human performance in different ways, for instance, compensating for decrease of ability, e.g. as a cognitive tool to remember tasks or guide in accomplishing activity [2, 12, 21]. Systems embedded in the environment can show the autonomous and proactive behavior, which typically is assigned computational programs denoted *intelligent and social agents*. If the software agents are dedicated to optimize human performance and well-being, they also need to detect, represent and evaluate activities performed by the human. Human activities however, are sometimes overlapping in time, and conducted in different order, utilizing different sets of sub-actions, which adds complexity to be handled. Moreover, a large part of human activity is managed through complex cognitive functions such as planning, reasoning, decision making, and developing knowledge and skills. In addition, human motivation and emotional influence on activity is significant, and is largely ignored in research on activity recognition.

Typically, methods for detecting human activities in daily living and working include statistical methods such as machine learning, where feature extraction and activity classification based on large amount of data is done (e.g. [5, 19]). We aim to address the activity recognition problem by using a semantic approach, based on theories of human purposeful activity. Another difference in our approach to activity recognition is that we adopt an activity-centered perspective instead of a user-centered perspective, where the user is included as one part of the activity system. This enables us to i) characterize the conduction of an activity, ii) relating this to the characteristics of the actor (capacities and limitations), and iii) to the contextual constraints on activity (domain knowledge, environment, etc.). We expect that this will facilitate the personalization of the support that an active assistive technology can provide [12].

The aim of the work presented in this paper is to develop a generic computer-interpretable model related to activity recognition and support generation based on informal and formal models for describing and explaining human activity (e.g., [11]).

The main contribution is a common model for representing the events generated when a human actor conducts activity, which captures the minimal information needed for advanced analyses of activity progression, performance and fulfillment of the human agent's goals and motives. This common model of activity can be used as base also for the software agent's agency, in order to define its purpose and goals.

The paper is organized as follows. A description of the methods applied is provided in the following section. The characteristics of activity is described in Section 3, followed by how representing activity as events (Section 4). In Section 5 we describe how activity can be detected based on the instruments a person uses. The results are discussed in relation to current research in Section 6 and further discussed in Section 7. The article ends with some conclusions and directions for future work.

2 METHODS

For creating a model of human activity and agency, theories and models of *human activity* developed within different research domains were studied. We primarily used the Activity Theory for categorizing and interpreting activities to represent knowledge about recognizable purposeful activities [11], and the International Classification of Functions, Disability and Health (ICF)¹ formed a baseline model for human activities and abilities.

We focus on three different knowledge domains related to health, which are relevant to a human actor who is an older adult. The *knowledge domains* are i) the diagnosis and management of neurocognitive disorders including dementia, ii) an intervention targeting older adults, related to the physiotherapeutic domain, and iii) older adults living in their home environment with some difficulties, representing the occupational therapy and nursing domains.

The model was implemented as two ontologies using Web Ontology Language (OWL)² which is an extension of the Resource Description Framework (RDF)³. RDF is a graph-based data model with labeled nodes and directed, labeled edges, which makes it a flexible and dynamic model for representing data. The nodes and edges can be augmented with additional information, and the edges represent the relationship between two resources. The Web Ontology Language (OWL) is an international standard for encoding and exchanging ontologies (i.e., terminology models) and is designed to support the Semantic Web. Consequently, the structures can be used for reasoning. RDF and OWL are the underlying representation formats for the models developed as a part of this paper.

¹<http://www.who.int/classifications/icf/en/>

²<http://www.w3.org/OWL/>

³<http://www.w3.org/RDF/>

The model is integrated in applications for dementia care, falls prevention and an ambient assisted living environment.

3 CHARACTERISTICS OF ACTIVITY CONDUCTION

Following Activity theory, an activity is defined by its objective and consists of a set of goal-oriented actions. An *activity system* includes the actor or actors, who are using instruments and the instruments are viewed as *mediators* for activity [11]. The role of objects can be either *instruments*, i.e., mediators, or the focus of activity, i.e., the *objective of activity*. The role of an object often shifts, by the actor's shifts of focus. We will in this section outline a model, which implements these activity-theoretical concepts and their relationships. Detecting *focus shifts* and the *role of objects* are considered a means for detecting objectives and goals for activity, and consequently, providing support tailored to the human actor's reasoning process and style of thinking. It should be noted that an object can be either a physical item or mental, abstract artifact. It should also be noted that in our model we include both human and software agents as actors.

Assistive agents need to detect human activities of different kind in order to provide appropriate and situated support. The kind of support to be provided can be adapted to the different ways in which an activity is conducted. An assistive computer application can be more or less structured and proactive, depending on the needs and preferences of the human actor. For example, the dementia diagnostic support may function as a template, or a checklist in the data capture and diagnostic process following a forward chaining process (i.e., diagnostic reasoning). The support may also be provided following a more causal reasoning process, where the human actor can propose a hypothesis, and have an argumentative dialogue with the application agent about the strength of the hypothesis (e.g. [25]). These two ways represent the same activity, since they share the same motive and will have the same outcome: an optimally assessed diagnosis.

However, these two ways to conduct the same activity are different, and become represented differently in the *event repository* defined and implemented in our work. While the main activity in both cases is dementia investigation, the sub-actions and their organization will differ, which will follow the individual physician's line of thinking. Moreover, both contain reasoning and decision making, but only one is *social*, in the sense that negotiation between two potentially conflicting standpoints is included.

Our third example case is when an activity is carried out in a physical, or physical-virtual environment where the physical and virtual items play the role as instruments. In this situation, skills and knowledge about how using the

physical and virtual tools and environment are needed. This kind of knowledge is different from the knowledge and skills in a particular medical knowledge domain required in the dementia example, and the social and behavioral knowledge and skills required in the dialogue example.

Table 1: Activities categorised by the type of instruments mediating the activity

Activity constraint	Required skills	Instruments	Example
Activity requiring domain knowledge	Factual and process knowledge in a particular domain	Structured knowledge-based protocols	Activities based on evidence-based clinical diagnostic criteria
Activity requiring social knowledge	Social knowledge	Language, dialogue and behavioral norms	Human-agent dialogue activities
Activity depending on a physical(-virtual) environment	Handling physical environment	Physical instruments available in the environment	Physical activities (body and space) or Instrumental Activities of Daily Living (iADL) such as maintaining food and medication intake routines

In the following sections, we categorize types of activity following these three categories of knowledge and skills required to conduct the activity (Table 1). This way we can assess the capacity of the human in performing the activity, based on how they handle the instruments. Moreover, by representing activity and utilize a purposeful model of the activity in the provision of support to the human, higher-level goals and motives can be inferred by their actions and interactions with the supporting agent.

4 REPRESENTING THE HUMAN ACTOR'S ACTIVITY AS OBSERVED EVENTS

The basic concept commonly used in generic ontologies is *event*, for representing an observable happening in the world[18].

We chose to build the *Actor repository* based on an event class, where each instance of an event has properties that

relate the event instance to information about the event (Figure 1).

In our implementation an event has a *start time* and an *end time* (Figure 1). End time means "end of focus time" following activity theory, meaning that the activity could either be ended, paused, cancelled or be ongoing in parallel, then requiring minimal attention.

Moreover, an event is always associated to an *activity*. The identifier of the activity is drawn from the *Domain knowledge repository* for the particular application. The activity associated to an event corresponds to one of the three types of activity introduced in the previous section: 1) domain knowledge-based (protocol-based), 2) social knowledge-based (dialogue-based), or 3) physical environment/instrument-based, observed in a physical environment.

Each event may also have associated *observations* and *evidence* (Figure 1). The difference between these two types of information is that evidence is information, which has been preceded by some reasoning, either by a human or software agent. As such, it can be challenged, and defeated in an argumentative reasoning task, e.g. in a dialogue. Moreover, evidence can be treated as the outcome of an activity, e.g. as a decision about a diagnosis in the dementia example.

An observation can also be challenged by conflicting observations that may contradict the first observation.

The events represented in the Actor repository are typically nested, which represents the Activity-theoretical hierarchical model of activity where human activity contains several layers of actions.

5 DETECTING ACTIVITY THROUGH THE INSTRUMENTS USED

In this section the three contextual constraints on an activity that relate to instruments are further explored and exemplified. We begin with the physical, or physical-virtual environment.

Activity Constrained Primarily by a Physical Environment

A fundamental view of Activity Theory is that human activity is mediated through *instruments*, or tools [11], which is embedded in the concept iADL (*instrumental* activities of daily living). Consequently, the primary key to identifying activity is to detect the instruments involved in the activity. A human actor uses both her body and other objects as instruments. One example is that based on 3D sensor information, the human movement of an arm and hand with an object in the hand can be interpreted as an activity involving the object as the instrument, while a movement without an object may be interpreted as a gesture with some communicative purpose, where the arm and hand become instruments.

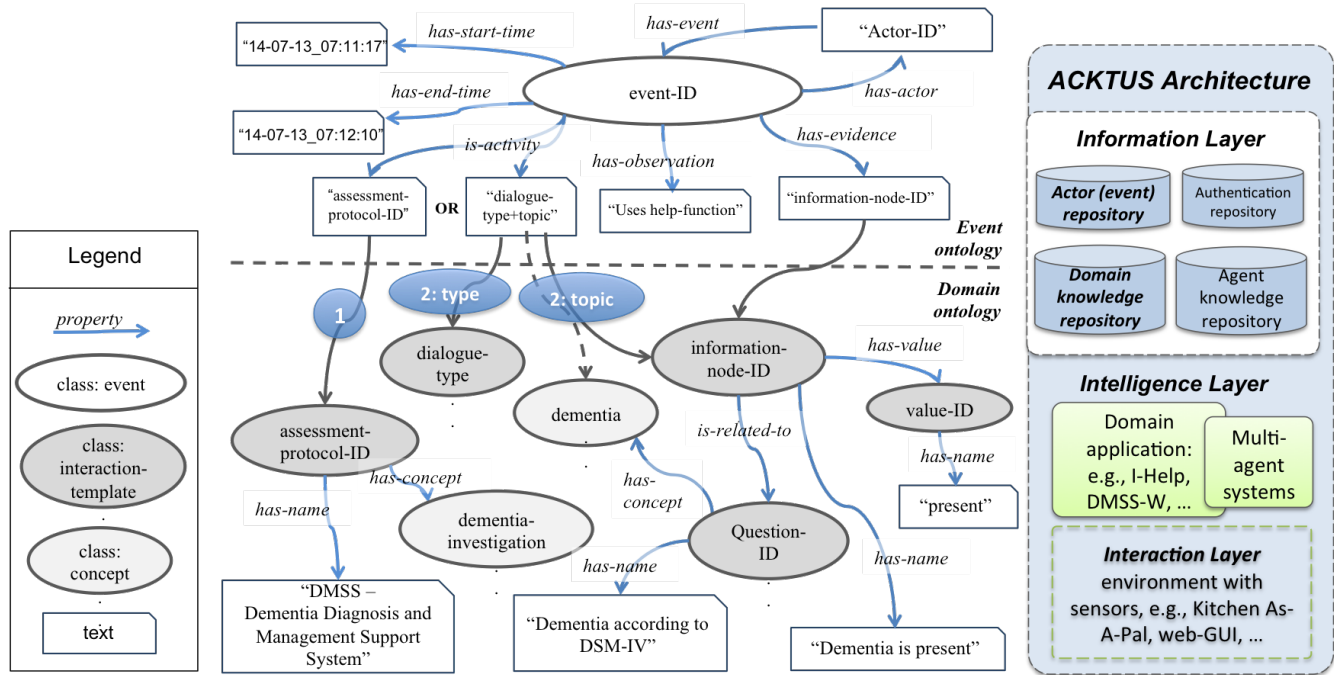


Figure 1: Example from the dementia domain where Number 1 exemplifies the *protocol-based activity*. Number 2 exemplifies *dialogue-based activity* where depending on dialogue type it can have as topic a concept, e.g., *dementia*, or an information-node that is a statement e.g., *Dementia is present*.

Therefore, next the *role of the involved object* needs to be determined. There are two options: the object may be the focus of activity, or merely a mediator of activity. If the object is the focus of activity, this reveals the objective of the activity. One example is when a person is cooking porridge, there are objects involved such as the pan, spoon to stir the content, milk, water, oat meal and salt. The focus of activity is what is becoming food in the pan, which can be detected by the amount of visual focus is given to the cooking pan, and not the salt, oat meal, milk or water.

The collected data obtained by the agent in an environment are interpreted into *observations* at a low level. These are typically collected using various sensors, such as accelerometers, 3D sensors etc. and can be operations such as gestures, holding objects, moving, sitting still, clicking on a particular button in an application, etc. Such observations are associated to an event by the property *has-observation* (Figures 1, 2). When these observations are obtained in a physical environment, the agent analyses the situation, and associates the set of observations to a preferred activity pointed out by the human actor during a baseline assessment, and which is represented in the domain ontology, e.g., "having breakfast" or "taking medication".

Domain knowledge-based activity

The domain ontology for each knowledge domain and related application has a class for *interaction templates*, which represent the instruments which the domain professionals use in their design of a support application, when using the knowledge-management system ACKTUS. One sub-class is *assessment-protocol*, which represents templates for conducting activity [15]. All assistive technology applications developed using the ACKTUS platform are based on an instance of this class, which contains other assessment-protocols representing composite sub-actions, or content such as questions with typically structured answer alternatives [14, 15]. These protocols can be interrelated and be executed as a kind of a loosely coupled workflow in order to aid the user in accomplishing a task. In this way, they represent semi-dynamic *hierarchies of activities*, following the notion of activity theory.

A *protocol-guided activity* is identified by the assessment protocol's associated activity concept, and the assessment protocol itself functions as an instrument in the performance of the activity. For instance, the assessment protocol building up the dementia support application has the name *Dementia Diagnosis and Management Support System*, and has the concept *dementia investigation* (Figure 1). Consequently, the name of the tool in combination with the concept, pointing

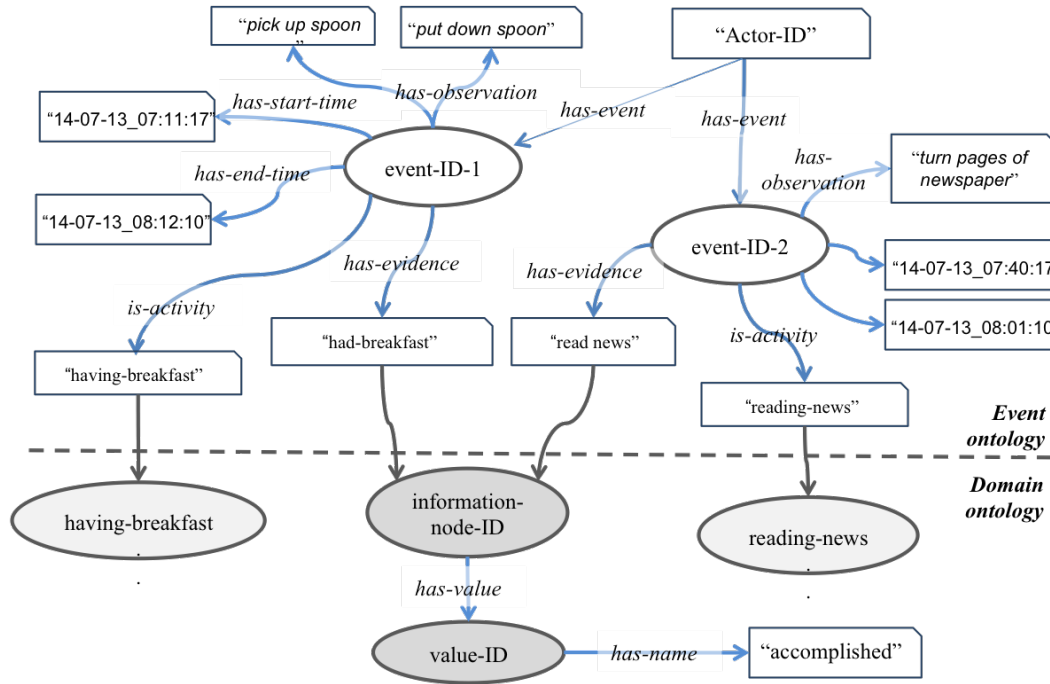


Figure 2: Example from a physical home environment where an older adult is having breakfast while reading the news. The identifier of the event is a concept that represents an activity.

to which activity the tool is aimed for, follows the activity-theoretical model of purposeful activity, which is necessarily mediated through instruments.

Social knowledge-based activity

Activity that requires social knowledge is for example, dialogues between an older adult and a nurse on health-related topics. In the case when an intelligent and social software agent acts as a knowledgeable dialogue partner, it also needs to behave in a way, sensible to the human and follow the human's line of reasoning across topics. In our approach, we define such *human-agent dialogue* by the *generic goal*. This generic goal is related to the type of dialogue that is chosen, where the type refers to the purpose and expected outcome of the dialogue. One example is *inquiry dialogues*, which aim at creating new knowledge in collaboration between the actors taking part in the dialogue [24]. During the dialogue however, the human and agent interact over one or more sub-dialogues on a *specific topic*. For example, when the older adult meets a nurse, the nurse has certain information about the person through the patient health record system (EHR). In our human-agent dialogue activity, the agent fetches this information about the human from the Actor(event) repository. The agent accesses the previous event-ID, and gathers information about the last dialogue if any with the person based on the Actor-ID. If the person has

initiated the dialogue, then the generic goal is based on the specific topic, such as "memory" which means the generic goal is to figure out if memory-related problem is being faced by the person.

Continuing with the example, the nurse asks memory-related questions to the human, if the human is finding "difficulty in doing everyday activities", takes any "medication" etc. Once the nurse has information about the human's health condition, she can make a conclusion about the presence/absence of memory problem and together they come up with solutions of "what can be done to reduce the memory related problem" being faced if any. In the human-agent dialogue activity, the agent provides advice to the human on what can be done for the betterment of the condition. Sometimes it can mean to "change priority of some activity" by the human, such as taking medication to be of highest priority than previously being only important. Another example is when the agent fetches previous event-ID and observes that the human has not called her son, then the agent can try to persuade her to call and talk with her son. The software agent follows certain dialogue and behavioral norms based on the preferences and priorities selected by the human to decide when, where and how to present the information. For example, the agent performs an activity on behalf of the human which the human has decided to do, such as, "send reminders about an appointment with the nurse". It is to

be noted here that all the text in quotation marks forms a sub-goal for the agent during a dialogue. The human and agent switch between different types of dialogues within a generic goal with topic: memory. Thus, a human-agent dialogue starting with a particular topic, can be nested to meet sub-goals in the process of achieving the overall motive for the dialogue activity. In this way, the activity-theoretical notion of hierarchy of activity can be represented in a dynamic and situated way, following the human's cognitive processes.

6 RELATED WORK

Recommender systems are widely used in many Internet-based programs [1, 17], where the user suffers from "information glut" and it is difficult to find the correct data he/she wants. In such systems a user profile is typically generated based on the user's historic click behavior. With this use profile, the system can predict which kind of data the user may be interested in and then select the data accordingly, providing a personalized version of the content to this user to save the time otherwise spent on information searching. The purpose can be to help finding the preferred information (i.e., optimizing activity performance), or present advertisements, which may trigger the person to buy a product (i.e. persuasive, encourage a kind of behavior change). A persuasive recommender system for social networks based on argumentation is presented in [4]. The agent in the dementia application exemplified in this article has similar functionality as recommender systems, in that it detects activity by click events. However, considering the semantically enriched information, which can be derived from the click events when interpreting the events through the ontology, our approach moves beyond traditional recommender systems. The interpretations done by the agent can also supplement the agent representing the expert physician in the application, by providing arguments for why the physician may have reasons to change his diagnostic routines [16].

Other agent-based systems have been developed for detecting activity with the goal to optimize activity performance (e.g. [9, 10, 22]). SHARE-IT is a project, which proposes the integration of multi-agent systems with other technologies to build e-tools for people with disabilities and for senior citizens. The tutorial application iTutorial has been developed as part of the assisted living environment in the SHARE-IT project [21]. Based on physical location and a set of individual-specific features (e.g., presence of memory impairment) the support is given in the form of detailed step-by-step instructions on how getting dressed, etc. The support relies on a comparable simple model of the senior citizen, where dysfunction is directly corresponding to type of support to be provided in pre-defined activities of

daily living, without adapting to variations of performance. The actor and activity ontology in our work is partly based on a holistic model of ability, dysfunction and health provided by the World Health Organization, which is familiar to health professionals. This differentiates our ontology from e.g., the semantic user model presented in [10], which separates the user ontology from the environment ontology, and matches the two for the purpose to identify handicap situations, mainly due to constraints in the physical environment. Another example of an automated system for older adults with cognitive impairment is presented in [20], which focuses mainly on generating reminders about the activities of their daily living. The approach presented by Erriquez and Grasso serves the individual with tailored advice, also based on predefined user models [7].

7 DISCUSSION

The categorization of activities described in this paper provides an outline for how to represent, detect and interpret information in order to assess activities meaningful to an individual. It has been found valuable to distinguish between these categories, since different methods are required for detecting these different activities, for detecting how a person is learning to master the instruments, and for providing support.

It has also been found valuable to combine these three categories of activity for providing tailored support to humans in their conduction of activity.

Different multi-agent systems have been developed and been integrated in different kinds of support applications: the dementia diagnosis and management system (DMSS-W) [25], I-Help, which is a mashup application for supporting older adults [3, 25] and Safe Step/Balansera [13]. DMSS-W and I-Help are two examples where the protocol-based activity is combined with the dialogue-based activity (Figure 1). 3D sensors among other sensors are collecting raw data for activity recognition in the Ambient Assisted Living (AAL) environment Kitchen As-A-Pal [23] (Figure 2) and Balansera [8]. The data is analyzed for detecting movements and objects in order to associate these to activities defined in the domain ontology. The information is subsequently used for analysis and provision of support by the I-Help or Safe Step applications.

The dialogues between the human and software agent serves as a followup of how the human thinks that their daily life works and their current emotional as well as health state. This subjective view is to be supplemented in our work with the above mentioned activity recognition system in the environment. Thus, we view the human-agent dialogues as a means to provide the human control over the AAL environment.

8 CONCLUSIONS

This paper provides a computer-interpretable model for detecting and representing activity in human-agent interaction, which is based on informal models for describing and explaining human activity and earlier research on intelligent agents for promoting wellbeing and health through supporting humans' activities. The results include a common model for storing events generated when a human actor conducts activity, which captures the minimal information needed for advanced analyses of activity progression, performance and fulfillment of the human agent's goals and motives. The model is implemented in different prototype applications targeting the activities in focus for our study.

Ongoing and future work include designing the dialogues and the pro-active person-tailored supportive behavior of the dementia application and the I-Help application. The dementia application is currently being evaluated in clinical practice, and feasibility studies will be conducted also of the other applications within the near future.

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